Exploring Camera Traps Image Classification: An Approach Using Supervised and Self-Supervised Learning with Deployment in Gradio Framework

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Abstract

The image classification of species in camera trap images presents significant opportunities for biodiversity monitoring in areas with many endangered animals. However, it has also remained challenging for researchers due to its high labor and time costs. This report examines the effectiveness of Convolutional Neural Networks (CNNs) for camera trap image classification, specifically with smaller to medium-sized datasets, employing six different supervised learning models alongside a self-supervised learning model. Each model was trained on a subset of 8594 images with 11 total species from a Florida wildlife camera trap study to determine if comparable accuracy could be achieved with 8.2% of the total data. The performance of the models was evaluated based on standard metrics such as accuracy. Our Graphics Processing Unit (GPU) computational power was one of the limitations that led to extremely long training periods for models and a low number of training epochs per model. Despite the limitation, our analysis shows supervised CNNs consistently perform in image classification, achieving 49-63% test accuracy after 5 epochs. Self-supervised learning was explored but deferred due to GPU constraints. Also, this study highlights the adaptability of CNNs to environmental data analysis and their potential to significantly advance the field of automated wildlife surveillance.

1. Introduction

This report explores how image classification can improve biodiversity monitoring by analyzing wildlife images obtained from camera traps. The vast volume of data generated by these traps and labeling them poses significant challenges in terms of processing time and labor intensity.

Outline Section 1 provides an introduction to fundamental concepts such as Camera Trap devices, Image Classification, and CNNs. Section 2 presents an overview of previous research related to the project topic. Sections 3, 4, and 5 detail the methods, dataset, and experiments conducted in the project. Sections 6, 7, and 8 present the results, discussion, and conclusions drawn from the conducted experiments.

Figure 1. Camera Trap deployed in the wild [12].

1.1. Camera Traps

Camera traps, often utilized in wildlife research and conservation, consist of a motion sensor and a camera. Positioned strategically, these devices automatically capture images or videos of wildlife as they pass by, enabling the collection of valuable data on behavior, population dynamics, and habitat usage without causing disturbance [12]. Activated by motion or heat, they are particularly effective for observing nocturnal or elusive species such as lizards or spiders. The introduction of camera traps has revolutionized ecological studies by facilitating the acquisition of extensive data on animal behavior, population, and distribution, particularly benefiting endangered species. Their deployment in remote or inaccessible areas also allows researchers to monitor wildlife dynamics over extended periods.

1.2. Image Classification

In wildlife monitoring, image classification is the act of assigning a label to an image according to its visual characteristics [14]. This is particularly important for identifying species captured in camera trap images. Traditionally, experts manually review these images, a process that is time-consuming and susceptible to errors, particularly with large datasets. Automated image classification using machine learning provides a quicker and more precise alternative, allowing for the efficient processing of numerous images [13]. This enhances the usability of data for conservation and ecological research purposes.

1.3. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a type of deep learning algorithm particularly effective for processing images. Convolutional Layers execute automatic feature extraction which allows the detection and interpretation of visual cues such as edges, textures, and complex patterns. In ecological research specifically, CNNs are employed to automate the identification of species in camera trap images, significantly reducing the workload and improving the accuracy of species classification and biodiversity assessments. These models have seen huge successes in recent years, as one 2018 study demonstrated that, in this context, CNNs reduce manual labor by 98.2% while maintaining an accuracy level of 96.6% [1].

2. Related Work

Past approaches in implementing CNN architectures in this domain include employing FasterRCNN and Inception-ResNet as the feature extractor by Christian et al. in 2020; using VGG and ensemble models for the separate tasks of animal detection and classification by Norouzzadeh et al. in 2019; and deploying ResNet-18 to achieve an unprecedented 97.6% and 99.9% top-one and top-five accuracy on author-constructed datasets by Tabak et al. [18].

Another effective method is to integrate multiple pre-trained CNN models to improve image classification. Margarita Favorskaya and Andrey Pakhirka developed a VGG-based model with two VGG16 branches and one VGG19 branch [4]. The VGG16 branches detect the muzzle and a part of the animal object respectively, and the VGG19 branch detects the entire animal object [4]. Top-1 and Top-5 percent accuracies were 80.6% and 94.1% for balanced data, whereas the unbalanced data accuracies were 38.7% and 54.8% [4]. Although the joint VGG model performs well on balanced data, two major drawbacks are that false positives and miss rates for the balanced data are quite large and range anywhere from 6.5%-24.8% across the species studied and that model training accuracies are poor for unbalanced data [4].

More recently, a 2020 study by Stefan Schneider and Saul Greenberg [14] trained six deep-learning computer vision network models on camera trap images using small training datasets and compared their relative performance. These six models were DenseNet201, Inception-ResNet-V3, InceptionV3, NASNetMobile, MobileNetV2, and Xception, and each model utilized transfer learning and augmentation techniques [14]. Due to the scarce availability of large, labeled wildlife image datasets and the tendency of CNNs to over-rely on scenery and the captured environment rather than the observed animal in an image, models trained on covert camera image data often suffer from overfitting [13]. The use of transfer learning with CNN not only removes the requirement of training on large, labeled datasets for attaining an acceptable training accuracy but also increases model generalization [3]. Similarly, data augmentation also increases model generalization. Our project employs the same method where six pre-trained models using transfer learning are trained using data augmentation techniques and compared for performance. In this project, we aim to employ five models outlined in the paper [14] and MNasNet 100 in place of NASNetMobile. We utilize these models on datasets of small to medium sizes and endeavor to compare the outcomes with those reported in the paper for similar datasets.

Another significant study pertinent to our project is the 2019 research conducted by Dan Hendrycks and Mantas Mazeika. Their study explored the application of self-supervised learning to generate representations for downstream tasks without requiring labeled data. The findings revealed that self-supervised learning outperforms fully supervised methods and is often regarded as a means to minimize the need for annotations [5]. In our project, we employ a pretext task involving background removal and color jitter data augmentation. We utilize the self-supervised learning model known as the Simple Framework for Contrastive Learning of Visual Representations (SimCLR) and attempt to train it on unlabeled data sourced from the internet, focusing on the same set of species used for supervised learning.

3. Dataset

Our project employs the labeled Florida Wildlife Camera Trap Dataset for supervised learning, encompassing wildlife images from two Florida locations: Corkscrew Swamp (Corkscrew) and Okaloacoochee Slough State Forest (OKSSF). The complete dataset comprises a total of 104,495 images across 22 species/classes. However, we utilize only 8.2% of the dataset, amounting to 8,594 images representing 11 unique species: Boar, Alligators, Opossum, Armadillos, Panthers, Bears, Raccoons, Coyotes, Bobcats, Otters, and Dogs. This selection focuses on small or medium-sized datasets. The subset includes both daytime and nighttime images. Additionally, for the unlabeled

dataset, we collected images of the same 11 species from the internet, resulting in a total of 9,719 images.

3.1. Software

Our project uses Google Colab Pro for supervised and self-supervised methods and deployment.

4. Proposed Method

We delve into supervised learning techniques and explore the relatively unexplored realm of self-supervised learning for classifying climate change and camera trap images. Our approach involves initially implementing traditional supervised learning algorithms and then looking into self-supervised learning techniques. This includes utilizing different pretext tasks deemed beneficial for image classification and extracting valuable feature representations from unlabeled data using various self-supervised learning-based feature representation models. We're conducting supervised and self-supervised learning to gauge research progress on small to medium-sized datasets. While supervised learning provides insights into existing methodologies, self-supervised learning remains unexplored in this domain. The abundance of unlabeled data prompts us to incorporate it into the project.

4.1. Supervised Learning

Supervised learning involves training models on labeled data to predict outcomes accurately. Our project involves training the following six supervised learning models and comparing their performance on the dataset we used: Inception V3 [15], DenseNet 201 [7], Inception-ResNet V2 [16], MobileNet V3 [6], Xception [2], MnasNet 100, where Mnas refers to Automated Mobile Neural Architecture Search [17]. By utilizing these models previously employed in camera trap datasets of small or medium sizes, we aim to interpret and comprehend the implications of image classification results for wildlife species.

4.1.1 Gradio

Gradio, a Python library, simplifies user interface creation for machine learning models. Developers can swiftly deploy interactive web interfaces without web development expertise. Users engage with models via intuitive interfaces, like sliders and dropdown menus. Our project employs the best-supervised learning model. The Gradio app predicts species from input images.

4.2. Self-supervised Learning

Self-supervised learning enables models to grasp and represent data without explicit labels, proving valuable when labeled data is scarce or expensive. The following subsections will elaborate on the algorithms used for this project.

4.2.1 Fast.ai

The Fast.ai library simplifies the extraction of internet data for unlabeled datasets, offering robust tools that streamline the process. Utilizing methods such as web scraping, API integration, and automated downloads facilitates efficient data collection. In our project, we leveraged these capabilities to obtain species images from the web, organizing them into folders named after each species, mirroring the same set of species used for supervised learning earlier. This compilation of unlabeled data serves as a valuable resource for training self-supervised learning models. The unlabeled dataset comprises 9,719 images for 11 species. This is identical to the dataset utilized for supervised learning with an increased image count, which enhances training efficacy by providing the model with more unlabeled data.

4.2.2 Contrastive Learning

Within self-supervised learning, contrastive learning plays a pivotal role, guiding a model to discern data by contrasting similar and dissimilar pairs. By generating positive and negative pairs of augmented data, the model refines its discrimination within a feature space, thereby improving its ability to generalize to new tasks. Our project employed the SimCLR model, which aims to differentiate features from images of species and applies the concept of contrastive learning in conjunction with predefined pretext tasks [5]. Unlike the cross-entropy loss utilized in supervised learning models, in this case, we employ contrastive loss, which is defined as a function before model training.

4.2.3 Pretext Task

We employed background removal and color jitter data augmentation techniques as pretext tasks. Background removal aimed to isolate species information within the image, refining the model's focus on species-specific features. Meanwhile, color jitter introduced color variations, enhancing resilience against shifts. These tasks played a crucial role in training the model to extract valuable representations from unlabeled data, thereby enhancing its performance on subsequent tasks.

In this domain, the issue of generalization and variations between day and night images is common. Models often perform well in the areas they are trained in and during testing but struggle in unfamiliar areas. Night images, in particular, require extra attention as they may not have been encountered during training, posing challenges for accurate prediction.

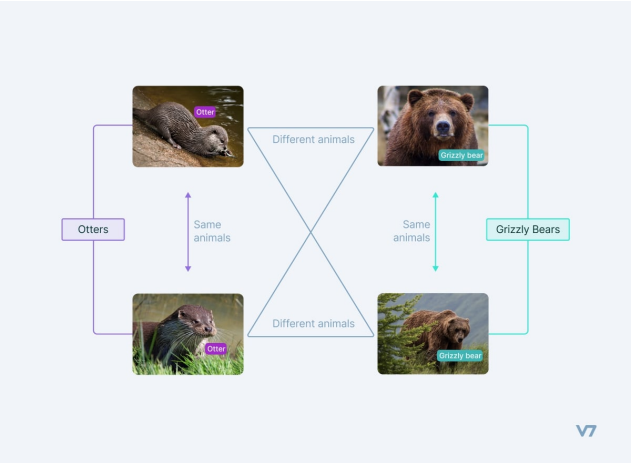

Figure 2. Contrastive Learning idea for species [8]

4.2.4 Models

We explored the Simple Framework for Contrastive Learning of Visual Representations (SimCLR), a self-supervised learning model that utilizes contrastive learning to learn data representations. This model trains a deep neural network to distinguish between augmented views of the same images (positive pairs) and different images (negative pairs) by optimizing contrastive loss [5]. This approach yields intricate representations that can be adapted to various computer vision tasks, demonstrating high-level performance. In our project, we aim to leverage this model to gain insights into contrastive learning for species and assess its potential utility in this relatively unexplored domain.

5. Experiments

The experiment involved both supervised and self-supervised learning approaches. Initially, six supervised learning models—Inception V3, DenseNet 201, Inception-ResNet V2, MobileNet V3, Xception, and MnasNet 100—were trained for five epochs using the labeled dataset containing 11 species from the Florida Wildlife Camera Trap Dataset. Data augmentation techniques such as RandomHorizontalFlip and ColorJitter, which adjust contrast, saturation, brightness, and hue parameters, were applied. The training process utilized the AdamW optimizer [10], a CosineAnnealingLR scheduler [9], and Cross-Entropy loss [11]. Batch sizes and learning rates were adjusted for each model to optimize training accuracy. Precision-recall and accuracy metrics were then evaluated to identify top-performing models for further training for 15 epochs, taking into account GPU resource constraints. Additionally, images were resized to 28x28 dimensions to facilitate efficient training in the subsequent 15-epoch models. Self-supervised learning using SimCLR was also explored, incorporating background removal and color jitter data augmentation techniques as pretext tasks. This self-supervised model was trained for 5 epochs using unlabeled data.

6. Results

Results for the supervised learning method were obtained for six models for 5 epochs, as presented in Table 1 and Table 2.

Model Name	Training Accuracy	Validation Accuracy
Inception V3	88%	85%
DenseNet 201	78%	69%
Inception-ResNet V2	95%	92%
MobileNet V3	95%	82%
Xception	97%	90%
MNasNet 100	96%	93%

Table 1. Training and Validation Accuracy of Six Models

Model Name	Test Accuracy
Inception V3	57%
DenseNet 201	49%
Inception-ResNet V2	62%
MobileNet V3	58%
Xception	63%
MNasNet 100	62%

Table 2. Test Accuracy of Six Models

Table 2 indicates that the test accuracy of the Inception-ResNet V2, Xception, and MNasNet 100 models is quite similar. Consequently, following the methodology outlined in the Experiment Section, we proceeded to train all three models for 15 epochs. Among these models, MNasNet 100 demonstrated the highest test accuracy, achieving approximately 50%. However, we made further adjustments to the hyperparameters compared to those used during the initial 5 epochs. Table 3 and Table 4 present precision, recall, F1-Score, and Support metrics, along with the number of test images, specifically for the best-performing model, MNasNet 100. Despite having fewer than 500 images for each of the 11 species, Table 4 reveals promising F1-Scores for at least 4 species. These findings suggest that our results are approaching those typically observed in research, where F1-Scores of around 75% are often attained for each species [14] with less computation resources.

The results of self-supervised learning can be seen in Figure 3 and the deployment of the best model in Gradio is shown in Figure 4.

Initially, the self-supervised learning training was conducted for 5 epochs. However, due to limited GPU resources, we were unable to complete even a single epoch and did not observe any contrastive loss, despite training for 5-6 hours and resizing the images to 28*28. Consequently, we provide an evidence-based training process in Figure 3.

Model Name	Precision	Recall
Alligators	0	0
Armadillos	0	0
Bears	0.11	0.07
Boar	0	0
Bobcats	0.42	0.67
Coyotes	0	0
Dogs	0	0
Opossum	0.55	0.36
Otters	0	0
Panthers	0.57	0.66
Raccoons	0.37	0.17

Table 3. Precision and recall metrics for each species

Model Name	F1-Score	Support
Alligators	0	6
Armadillos	0	26
Bears	0.08	103
Boar	0	24
Bobcats	0.52	493
Coyotes	0	30
Dogs	0	40
Opossum	0.44	281
Otters	0	5
Panthers	0.61	512
Raccoons	0.23	204

Table 4. F1-Score metrics for each species

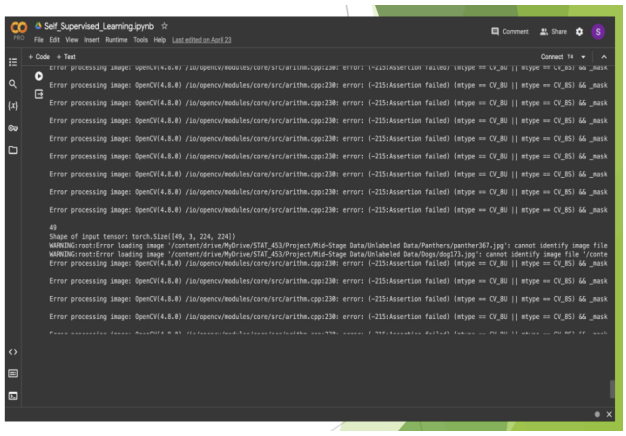

Figure 3. Training done for Self-Supervised Learning Method

7. Discussion

This study delved into assessing the efficacy of supervised and self-supervised learning methodologies in classifying camera trap images within the realm of wildlife monitoring. Our discoveries offer valuable perspectives on the strengths and weaknesses of contemporary deep learning methods in this field, shedding light on various directions for prospective investigation.

The supervised learning models, trained on labeled datasets, displayed diverse levels of effectiveness, with certain models surpassing others in both training and validation accuracies. Particularly, Inception-ResNet V2 and MNasNet 100 stood out by achieving remarkable training accuracies of 95% or higher, accompanied by validation accuracies ranging between 90% and 93%. These outcomes imply that deeper and more intricate architectures might excel in capturing the nuanced features inherent in wildlife images.

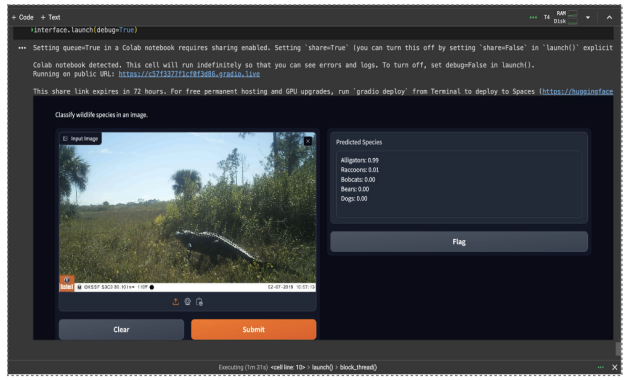

Figure 4. Deployment of the best model in Gradio

However, it's important to recognize that the performance of these models remained relatively modest, typically with validation accuracies ranging from 69% to 85%. This discrepancy between training and validation accuracies suggests potential overfitting, which could be mitigated through additional regularization techniques or by augmenting the size and diversity of the training dataset. Our findings align with prior research in this domain, which has also highlighted the difficulty of achieving high accuracies with small to medium-sized datasets. Studies conducted by Vecvanags et al. [18] and Schneider et al. [14] have similarly revealed the challenges faced by CNN models trained on camera trap images, particularly concerning overfitting and generalization issues when dealing with datasets featuring limited species diversity or environmental variability.

Numerous challenges and constraints were encountered throughout this study, potentially influencing the outcomes. A primary challenge stemmed from the limited availability of labeled data, restricting the diversity of species and environmental scenarios within the training dataset. This emphasizes the critical need for concerted data collection endeavors and interdisciplinary collaboration to construct comprehensive, annotated datasets for training and assessment purposes. Furthermore, the study faced constraints regarding computational resources allocated for model training, imposing limitations on training duration, batch size, and model intricacy.

Despite these obstacles, our study through the utilization of supervised and self-supervised learning methodologies, researchers can streamline image analysis, decrease manual workload, and extend monitoring initiatives over expansive geographical regions. Furthermore, by analyzing extensive

datasets of images captured by camera traps, researchers can glean valuable insights into ecosystem dynamics and species interrelationships. These insights can in turn inform the development of conservation strategies and policy decisions grounded in empirical evidence.

8. Future Directions

Moving forward, this study points to numerous directions for future research. Primarily, there is a need to enhance and diversify the labeled datasets utilized for training machine learning models, encompassing a wider spectrum of species, habitats, and environmental contexts. Collaborative endeavors and agreements for data sharing could expedite the process of dataset expansion and annotation by pooling together resources and expertise.

Moreover, it is imperative to prioritize research into innovative model architectures, transfer learning methodologies, and data augmentation strategies to enhance the effectiveness and resilience of machine learning models in camera trap image classification. Investigating self-supervised learning methods, including contrastive learning and pretext tasks, holds the potential for harnessing unlabeled data and mitigating challenges associated with data scarcity.

Lastly, through the integration of machine learning and data-driven methodologies, we have the opportunity to deepen our comprehension of biodiversity dynamics and play a pivotal role in safeguarding and sustainably managing ecosystems worldwide.

9. Conclusion

Our objective was to assess the feasibility of training a deep learning model to autonomously detect wildlife and categorize them into predefined classes using self-supervised learning, utilizing a limited set of manually labeled camera-trap images. In this study, we evaluated the efficacy of CNNs in automatically extracting pertinent information from images and assigning them to respective categories.

Our examination of camera trap image classification for species monitoring has unveiled the capabilities and constraints of contemporary machine learning methodologies. Through the utilization of supervised learning techniques, we have showcased the efficacy of Convolutional Neural Networks (CNNs) in automating species identification endeavors, albeit with modest levels of accuracy. Despite our top-performing models achieving approximately 50% accuracy following 15 epochs of training, this outcome does not meet the anticipated 75% accuracy benchmark for medium-sized datasets with analogous parameters, as observed in prior research [18].

Our findings stress the vital role of species monitoring through image classification, vital for conservation and species preservation. Additionally, classifying camera trap images aids in annotating extensive unlabeled data, advancing biodiversity research. However, challenges remain. The scarcity of labeled data prompts the exploration of alternative methodologies like self-supervised or semi-supervised learning. Moreover, models must generalize beyond their training environments, prompting inquiry into the efficacy of current techniques and the benefits of diverse datasets, including images from both day and night.

Considering these obstacles, our investigation into self-supervised learning emerges as a promising direction for future research. Through the utilization of unlabeled data for training, this methodology presents an innovative approach to species classification and serves as a means to overcome the constraints encountered by supervised learning models trained on limited datasets.

Looking ahead, it is imperative to persist in refining and innovating within the realm of camera trap image classification. By tackling these obstacles and embracing novel methodologies, we can propel the automation of wildlife monitoring forward and make meaningful contributions to the preservation of our planet's invaluable biodiversity.

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